

Climate 5000 Series - Gen 4

12K BTU Single Zone Regular System

Wall Mounted IDU

Engineering
 Submittal
 Sheet



BOSCH

Overview and Project Information



Project name	
Location	
Architect	
Engineer	
General contractor	
Reference #	

12K Btu Single Zone Regular System - Wall Mounted IDU

Indoor part number		8-733-962-673
Indoor model number		BMS500-AAU012-1AHWXD
Outdoor part number		8-733-962-685
Outdoor model number		BMS500-AAS012-1CSXRD
Efficiency Data		AHRI 210/240 2023
SEER2		25.5
EER2		13
HSPF2 Region IV		10.6
HSPF2 Region V		8.0
COP(47F)		3.90
COP (5F)		2.30
Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	2900~14500
	Rated Capacity	12000
	Current	4.1
	Input Power	916
Heating	Capacity Range	4500~15400
	Rated Capacity (47F)	12000
	Rated Capacity (5F)	8400
	Current	4.0
	Input Power	902
Electrical Data		
Power supply (V / Ø / Hz)		208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)		3.0 / 12.0
Max. fuse indoor/outdoor (A)		15.0 / 15.0
Compressor Data		
Type:		Rotary
Refrigerant oil /oil charge (ml)		VG74 / 310
Input (W)		325 / 826
Rated current (RLA/A)		2.40 / 5.65
Operation Range		
ODU	Cooling (°F / °C)	-13~122 / -25~50
	Heating (°F / °C)	-13~75 / -25~24
IDU	Cooling (°F / °C)	60~90 / 16~32
	Heating (°F / °C)	32~86 / 0~30

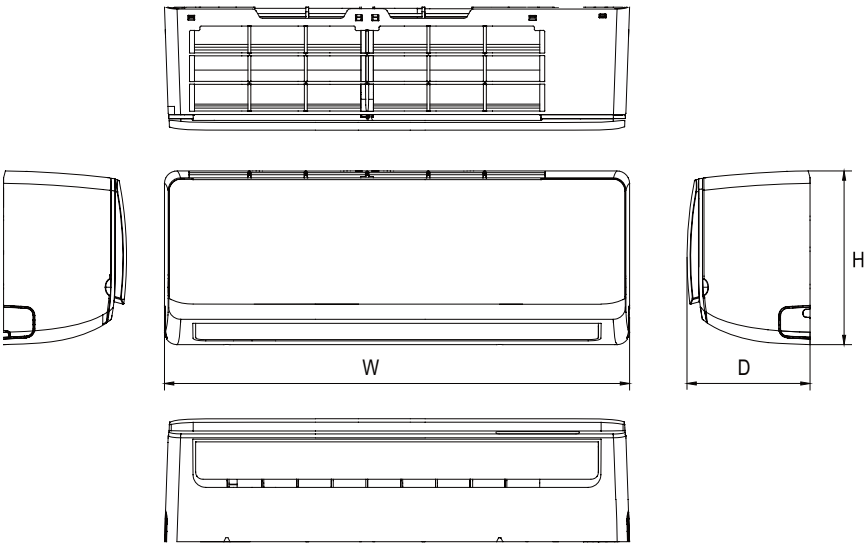
Piping

Refrigerant precharge (oz/kg)		32.45 / 0.92
Length for precharge (ft/m)		24.6 / 7.5
Liquid side (in/mm)		1/4" / Ø6.35
Gas side (in/mm)		3/8" / Ø9.52
Additional Charge of Refrigerant (oz/ft)/(g/m)		0.16 / 15
Max. refrigerant pipe length (ft/m)		82 / 25
Max. difference in height (ft/m)		49.2 / 15
System Data		
Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		2.32
IDU	Fan motor model	ZKFP-20-8-6-7
	Fan motor qty (ea)	1
	Fan motor input (W)	50.0
	Fan speed (RPM)	1100 / 850 / 700
	Air flow, Turbo / H / M / L / Si (CFM)	441.4 / 323.7 / 229.6 / 176.6 / 147.2
	Approx. Air Throw Range ft (m)	23.0 / 7
	Noise level, Turbo / H / M / L / Si (dB(A))	46 / 38 / 33.0 / 18.0 / 18
	Net weight (lbs / kg)	22.93 / 10.4
	Gross weight (lbs / kg)	29.32 / 13.3
ODU	Fan motor model	ZKFN-34-10-1L
	Fan motor qty (ea)	1
	Fan motor input (W)	50.0
	Fan motor speed, H / M / L (RPM)	800 / 700 / 600
	Air flow, Max (CFM)	1235.3
	Noise level, Max (dB(A))	56.0
	Net weight (lbs / kg)	62.17 / 28.2
	Gross weight (lbs / kg)	67.68 / 30.7
Accessories		
BMS-WT2-XXD (8733962707)		Wired Wall Thermostat
BMS-DRY-XXD (8733969263)		24V Interface
BMS-PK2-7X1 (8733951011)		25ft Lineset
Standard Parts		
8-733-969-088		Remote Control

Climate 5000 Series - Gen 4
12K Btu Single Zone Regular System - Wall Mounted IDU

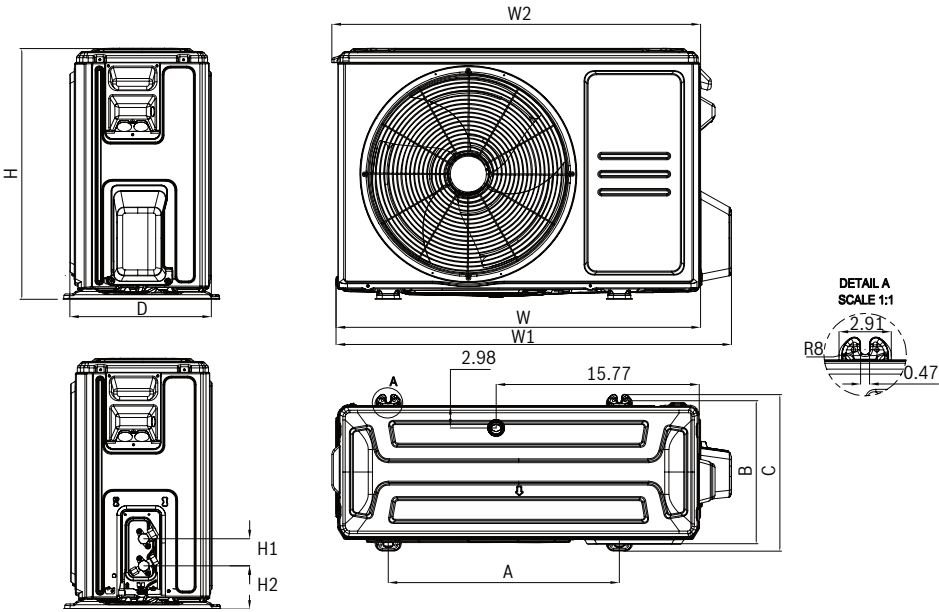


Indoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAU012-1AHWXD	31.42	9.21	11.85

Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	H1 (in)	H2 (in)	A (in)	B (in)	C (in)
BMS500-AAS012-1CSXRD	30.12	11.93	21.85	32.9	30.87	2.36	3.67	17.8	11.26	12.36

Climate 5000 Series - Gen 4

12K Btu Single Zone Regular System - Wall Mounted IDU



BOSCH

Heating Performance - Indoor Dry Bulb Temperature at 441 CFM										
Outdoor DB(°F)	TCITotal Heating Capacity(BTU/hr)					PIPower Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-13°F(-25.0°C)	5977	5857	5828	5804	5752	0.97	0.98	0.98	0.98	0.99
-4°F(-20.0°C)	7359	7211	7175	7147	7082	1.04	1.05	1.05	1.06	1.06
0°F(-17.8°C)	8034	7873	7834	7803	7732	1.08	1.09	1.09	1.09	1.10
5°F(-15.0°C)	8878	8701	8657	8623	8545	1.13	1.14	1.14	1.14	1.15
14°F(-10.0°C)	9931	9733	9684	9645	9558	1.18	1.19	1.20	1.20	1.20
17°F(-8.3°C)	10282	10077	10026	9986	9896	1.20	1.21	1.21	1.22	1.22
22°F(-5.6°C)	11060	10838	10784	10741	10644	1.26	1.27	1.28	1.28	1.28
27°F(-2.8°C)	11837	11600	11542	11496	11393	1.32	1.33	1.34	1.34	1.35
32°F(-0.0°C)	12614	12362	12300	12251	12141	1.38	1.40	1.40	1.40	1.41
35°F(1.7°C)	13081	12819	12755	12704	12590	1.42	1.43	1.44	1.44	1.44
37°F(2.8°C)	13527	13256	13190	13137	13019	1.40	1.41	1.41	1.41	1.42
42°F(5.6°C)	14642	14349	14277	14220	14092	1.33	1.35	1.35	1.35	1.36
44.6°F(7.0°C)	15088	14786	14712	14653	14522	1.31	1.32	1.32	1.33	1.33
47°F(8.3°C)	15757	15442	15365	15303	15165	1.27	1.28	1.28	1.29	1.29
52°F(11.1°C)	15683	15369	15292	15231	15094	1.18	1.19	1.19	1.19	1.20
57°F(13.9°C)	15609	15297	15220	15159	15023	1.08	1.09	1.09	1.09	1.10
59°F(15.0°C)	15579	15268	15191	15130	14994	1.04	1.05	1.05	1.05	1.06
62°F(16.7°C)	15535	15224	15148	15087	14951	0.98	0.99	0.99	1.00	1.00
64.4°F(18.0°C)	15499	15189	15113	15053	14917	0.94	0.95	0.95	0.95	0.95
69.8°F(21.0°C)	15419	15111	15035	14975	14840	0.83	0.84	0.84	0.84	0.85
75.2°F(24.0°C)	15339	15032	14957	14897	14763	0.73	0.74	0.74	0.74	0.74

Bolded values were tested. All other values are simulated and for reference only.